

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th November 2019

Subject SP5 – Investment and Finance

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** Outline the risks involved in infrastructure investments. [6]
- Q. 2)** Calculate the value of a swap option which gives the holder the right to enter into a four years annual pay swap in 2 years where a fixed rate of 6% p.a. is paid and floating rate is received. Assume the below spot rates for your calculation:

- 3-year spot rate: 6.50%
- 4-year spot rate: 6.60%
- 5-year spot rate: 6.75%
- 6-year spot rate: 7.00%

The swap principal is 100 crore, forward spot rate is 7% and the volatility of the swap rate is 25%. [8]

- Q. 3)** An insurance company launched a single premium unit-linked product, with five years to maturity, that offered capital guarantee i.e. maturity benefit is higher of the single premium or the maturity fund value. This is the first unit-linked product of the insurer that offers investment guarantees. A new fund was launched for this product. The product was open for subscription for three months and proved to be a great success with Assets Under Management (AUM) of the product accounting to 26% of the AUM of total unit-linked business. As on date, overall split for all unit-linked funds is 70% bonds and 30% equities. All old unit-linked funds are rebalanced quarterly to conform to their respective target allocations.

% AUM	Fund	Target allocations	
		Equities	Bonds
40%	Debt funds	0% to 10%	90% to 100%
24%	Balanced funds	25% to 35%	65% to 75%
10%	Equity funds	90% to 100%	0% to 10%
26%	New fund	0% to 100%	0% to 100%

Fund manager got net premiums i.e. 95% of single premium for investment, rest 5% spent for company expenses, agents' commissions and cost of cover. He initially invested net premiums 50% in equities and 50% in bonds in the new fund. After two days from close of subscription to the new fund, equity markets crashed by about 40% and the investment committee decided to move completely into bonds for the new fund.

The insurer can lend or borrow at 6.00% p.a. The use of equity derivatives is not allowed by regulations. Use of simple interest rate derivatives / swaps, bond forwards and bond options are allowed only for hedging purposes.

- i) Discuss how to carry out 'equities to bonds' switch, at minimum cost for the new fund. (4)
- ii) Discuss various approaches of liability hedging for the new fund. (8)
- [12]
- Q. 4)** You have recently joined a Bank as an intern. Your manager has asked you to update her on the Monetary Policy Statement and other press releases issued by Reserve Bank of India.
- i) In October 2019, in monetary policy statement, RBI has reduced the repo rate to 5.15%. Discuss the objective of rate cut on Indian economy including expected impact on balance of payments. (8)

- ii) RBI has mandated Banks to link their lending rates to retail customers and micro, small and medium enterprises (MSMEs) with an external interest rates benchmark. Explain the rationale behind this decision.

(4)

[12]

- Q. 5) In a developed economy, all investors divide their assets between an equity index fund and a bond index fund, each managed by specialist fund managers. All investors can borrow or lend at the risk-free rate of 6% p.a. The table below summarizes the performance of the two types of funds.

	Bond fund	Equity fund
Mean Return	8.10%	12.60%
Standard Deviation	15.00%	21.00%
Covariance of bond and equity returns	0.007875	

- i) In mean-variance portfolio theory, what do you mean by:

a) Efficient frontier

(1)

b) Optimal portfolio

(1)

Assume only two risky assets (bond fund and equity fund) are available for investment. Under Capital Asset Pricing Model (CAPM), the optimal portfolio split between the bond and equity funds is 23.54% and 76.46% respectively. Following table gives mean return and standard deviation for portfolios with differing proportions of bond and equity funds.

Composition		Portfolio	
Bond	Equity	Mean Return	Standard Deviation
100%	0%	8.10%	15.00%
95%	5%	8.33%	14.55%
90%	10%	8.55%	14.17%
85%	15%	8.78%	13.88%
80%	20%	9.00%	13.67%
75%	25%	9.23%	13.55%
60%	40%	9.90%	13.76%
50%	50%	10.35%	14.35%
40%	60%	10.80%	15.25%
30%	70%	11.25%	16.41%
23.54%	76.46%	11.54%	17.28%

- ii) Plot six points that lie on an efficient frontier of risky assets in return – standard deviation space. One of the six points should represent minimum variance portfolio.
- iii) Show that the market price of risk is 32.06%.
- iv) Calculate two risk adjusted return measures and comment on the performance of two managers X and Y (see table below). Which one of the two risk adjusted return measures is more appropriate? Why?

(7)

(1)

	Manager X	Manager Y
Mean Return	16%	22%
Standard Deviation	22%	38%

(6)
[16]

Q. 6) A pension fund hires two different fund managers. Manager A looks after the fixed interest investments. Manager B looks after the equity investments. Manager A would like to position his portfolio to capitalize on expectations concerning the following four factors that influence interest rates:

- changes in the level of yield curve
- changes in the slope of yield curve
- changes in yield spreads between sectors
- changes in yield of a particular instrument

i) Describe a strategy for each of the four factors that could be used to exploit Manager A's expectations about that factor. (5)

ii) Manager B wants to protect his portfolio from unexpected fall in equities in next month. Discuss two strategies, using equity index derivatives, to protect Manager B's portfolio value without losing upside potential. Suggest how to choose one of the two strategies. (5)

iii) Discuss the relative merits of assessing performance of the two Managers with respect to:

- a) Published bond / equity indices; and
- b) Competitor portfolios with a target of coming in top quartile i.e. beat 75th percentile.

Your answer should highlight the difficulties you may face including any deficiencies of the assessment method. (8)

iv) Suggest an alternative assessment method and discuss its relative merits. (4)
[22]

Q. 7) Mr. XYZ has started a tours and travels company couple of years back which organizes domestic group tours across India. Though the Company is yet to break-even, but it has been well appreciated by the customers and has organized 50 successful tours. He wants to expand his business to international tours. For financing the expansion budget, he approached a bank but couldn't secure a bank loan.

i) He read somewhere that Private debt can be an alternative to bank funding and has approached you to understand this in detail. Describe Private Debt. (6)

ii) He wants your help in deciding whether he should approach a private equity firm or try for private debt. Evaluate suitability of both the options for him. (8)

Mr. XYZ also wants to start an innovative app-based travel company. For launching this company, he has budgeted total capital requirement of INR 10 crore. He has INR 3 crore from his earlier savings. For remaining capital, he has decided to approach an angel investor.

- iii) Suggest two tax incentives along with reasons for promoting start-ups. (4)
- iv) He has come across a report highlighting very low success rate of start-ups and is confused whether to start this Company where he expects to earn very handsome return provided the start-up becomes successful or simply earn interest by investing his savings in a FD with a Bank. Describe the behaviour of Mr. XYZ based on the theory of behavioural finance. (6)

[24]
